TABREED DISTRICT COOLING GROUP PROJECT

Project title	TABREED DISTRICT COOLING GROUP PROJECT
Project ID	4519
Monitoring period	25-May-2023 to 31-July-2024
Original date of issue	17-September-2024
Most recent date of issue	14-April-2025
Version	01.6
VCS Standard Version	VCS Standard Version 4.7
Prepared by	Shashi Prakash Abu Dhabi Future Energy Company PJSC Masdar

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PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

National Central Cooling Company PJSC (TABREED) is implementing highly efficient district cooling project titled 'TABREED DISTRICT COOLING GROUP PROJECT (hereafter called "Project" or "Grouped Project")'. The project will reduce electricity consumption by providing highly efficient district cooling solution (air conditioning) to meet the cooling demand for residential and commercial consumers in the Abu Dhabi Emirates, United Arab Emirates (UAE). The highly efficient district cooling will displace the cooling from conventional/less efficient cooling system and reduce the electricity consumption which would result in reduction in greenhouse gas (GHG) emissions.

In the UAE air conditioning is necessity to survive. The penetration of cooling technology is almost 100%. Most of the residential and commercial consumers use window air conditioner or split air conditioner to meet the cooling demand. The commercial consumers have started shifting towards air cooled centralized cooling. These technologies are less efficient with respect to district cooling.

The project instance 1 is 9000 TR capacity project, a new district cooling plant to supply cooling to residential and commercial consumers in the Sea World, Yas Island Abu Dhabi, United Arab Emirates. The Project is displacing isolated and less efficient cooling technologies (air-cooled reciprocating chiller systems) and thereby reducing the consumption of electricity and corresponding Carbon Dioxide (CO2) emissions.

The project instance 2 is also 9000 TR capacity project, a new district cooling plant to supply cooling to residential and commercial consumers in the Saadiyat Island, Abu Dhabi, United Arab Emirates. This project instance has been commissioned on 3rd May 2024.

The emission reductions from the project activity during the first monitoring period is mentioned in the table below:

Period	Emission Reduction Project instance 1	Emission Reduction Project instance 2
25-May-2023 to 31-December 2023	11,016	0
01-January2024 to 31-July -2024	9,537	2,275
Total	**20,553**	**2,275**

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	12-July-2024 (Registration date)	*VCS*	KBS Certification Services Limited	N/A
Verification	From 25-May-2023- 31-July-2024	VCS	DNV Business Assurance India Pvt Ltd.	1.18 Years

1.3 Sectoral Scope and Project Type

Sectoral scope[1]	1. Energy (renewable/non-renewable)
Project activity type	Type II – Energy Efficiency Improvement Project

1.4 Project Proponent

Organization name	NATIONAL CENTRAL COOLING COMPANY PJSC
Contact person	Antonio Di Cecca
Title	Chief Operating Officer
Address	PO Box 29478, Abu Dhabi, United Arab Emirates
Telephone	T + 971 2 202 0400 Ext 434
Email	adicecca@tabreed.ae

1.5 Other Entities Involved in the Project

Not Applicable.

Organization name	
Role in the project	
Contact person	
Title	
Address	
Telephone	
Email	*Note: The email address domain must match that of the organization*

[1] *Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes*

1.6 Project Start Date

Project start date	25-May-2023
Justification	As per section 3.8 of the VCS Standard (v4.5), the project start date of a non-AFOLU project is the date on which the project began generating GHG emission reductions or removals. The project activity is a non-AFOLU project and the date of commissioning of first Project instance is 25-May-2023 (the date on which emission reduction started). Thus, the project start date is as 25-May-2023.

1.7 Project Crediting Period

Crediting period	☒ *Seven years, twice renewable* ☐ *Ten years, fixed* ☐ *Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)*
Start and end date of first or fixed crediting period	25-May-2023 to 24-May-2030

1.8 Project Location

The grouped project will be located in Abu Dhabi Emirates in United Arab Emirates. The grouped project covers Abu Dhabi Emirates boundary as presented below[2]:

Latitude: 22°40' N to 25°North

Longitude: 51° to 56° East

The project activity instance 1 is installed in Sea World Abu Dhabi with following coordinates:

Facility Name	Sea World District Cooling Plant
Facility Activity	District Cooling Plant
Plot Area	10099.95-meter square
Location	Yas Island Sector YS6
Plot number	P10
Latitude	24.483103977 N

[2] https://en.wikipedia.org/wiki/Emirate_of_Abu_Dhabi

Longitude	54.618530045 E

The project activity instance 2 is installed in Saadiyat Island Abu Dhabi with following coordinates:

Facility Name	Saadiyat island District Cooling Plant 1
Facility Activity	District Cooling Plant
Plot Area	3752 -meter square
Location	Saadiyat Island SDE 2
Plot number	P4
Latitude	24.5476944 N
Longitude	54.4541389 E

1.9 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	AM0117	AM0117: Introduction of a new district cooling system	2.0
Tool	Tool01	Tool for the demonstration and assessment of additionality	7.0
Tool	Tool03	Tool to calculate project or leakage CO2 emissions from fossil fuel combustion	3.0
Tool	Tool05	Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation	3.0
Tool	Tool07	Tool to calculate the emission factor for an electricity system	7.0
Tool	Tool11	Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period	03.0.1
Tool	Tool23	Additionality of first-of-its-kind project activities	3.0

1.10 Double Counting and Participation under Other GHG Programs

1.10.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.10.2 Registration in Other GHG Programs

Was the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

1.11 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.11.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.11.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.11.3 Supply Chain (Scope 3) Emissions

Do the project activities affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☒ Yes ☐ No

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☒ Yes ☐ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through the Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities].”

☒ Yes ☐ No

The statement is hosted on the Tabreed website https://www.tabreed.ae/sustainability/

1.12 Sustainable Development Contributions

The proposed Project Activity contributes to sustainable development in the United Arab Emirates by the following means:

Economic indicators

The Project Instance contributes to the

1. Conservation of energy sources through a more efficient way of generation and distribution of cooling energy, thus allowing for decreased electricity consumption per ton of cooling supplied.
2. Education and training of staff in the operation of the district cooling system. In addition, staff will be trained in the application of the international VCS standards of the Project Instance Activity monitoring and quality assurance / quality control.

Social indicators

The Project Instance contributes to the positive quality of life and benefits the local community by helping ensure a less energy intensive generation and supply of cooling energy.

Environmental indicators

The Project Instance

1. Will be reducing an estimated total of 22,828 tCO_2 of emission reductions during this monitoring period and therefore contributing to UN SDG 13 ‘Climate Action’.
2. Complies with all applicable national environmental regulations and standards and has no impact on air, water, soil or groundwater pollution.

Technical indicators

The Project Instance introduces a new and energy efficient technology and know-how into the United Arab Emirates, which can successfully be replicated to other cities within the region. The Project Instance will contribute to the increase in utilization and application know-how of this type of technology (district cooling).

Table 1: Sustainable Development Contributions

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime
Sequential row number	*SDG Target number*	*Number and text of SDG indicator or, if no official SDG indicator is applicable, user-defined indicator*	*Indicate the project's contribution to the SDG Indicator (implemented activities to increase or decrease)*	*Brief description of the quantifiable impact of the project's activities related to the SDG indicator, during the monitoring period.*	*Brief description of the cumulative quantifiable impact of the project's activities related to the SDG indicator, over the project lifetime.*
1)	8.3	The development of project contributes to various activities i.e. feasibility study, design reports, manufacturing, installation, commissioning and operation. In addition, the grouped project activity will also have short term employments during construction. PP will recruit more employees for in different time with a focus on hiring females. Hence, the grouped project will achieve SDG 8. Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all	Implemented activities to increase the jobs	The grouped project activity created a minimum of 3 operators' employments in construction and operation.	The grouped project activity created a minimum of 3 operators' employments in the monitoring period.

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime
2)	11.6	By installing energy-efficient district cooling plant, it helps to decrease the quantity of air pollutants that are emitted into the atmosphere and the demand for electricity generated from thermal power plant. Hence, the grouped project will achieve SDG 11. Make cities and human settlements inclusive, safe, resilient and sustainable	*Implemented activities to decrease the pollutants*	Reducing the emission of fine particulate matter via decreasing electricity demand from thermal power plants.	Reducing the emission of fine particulate matter via decreasing electricity demand from thermal power plants.
3)	13.0	By implementing the energy efficiency district cooling project, the grouped project would lower GHG emission through an estimated reduction of more than 22,828 tCO2e during the first monitoring period. Hence, the grouped project will achieve SDG 13. Take urgent action to combat climate change and its impacts.	Implemented activities to increase emission reduction	Reduce to total electricity usage demand in the project area to generate 22,828 tCO2e of emission reductions	Reduce to total electricity usage demand in the project area to generate approximately 22,828 tCO2e of emission reductions till 31-July-2024.

1.13 Commercially Sensitive Information

Not applicable. The grouped projects do not have any commercially sensitive information.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	The project is an energy efficiency project. The stakeholders are: 1. Employees 2. Government/Competent Authority a. Department of Energy b. Abu Dhabi Distribution Company (ADDC) c. Environmental Agency Abu Dhabi (EAD) d. Abu Dhabi Department of Municipalities and Transport (DMT) 3. Local Communities 4. Non-Governmental Orgainsation (NGO)/Community Based Organisation (CBO)
Legal or customary tenure/access rights	Not applicable
Stakeholder diversity and changes over time	The stakeholders are diverse, and no changes are expected over time.
Expected changes in well-being	Other than emission reductions no other changes in the well-being of stakeholders.

Location of stakeholders	Stakeholders are diverse and located in the emirates of Abu Dhabi.
Location of resources	Not applicable.

2.1.2 Stakeholder Consultation and Ongoing Communication

Ongoing consultation	Tabreed has a customer service charter (https://www.tabreed.ae/customer/) which provides the detailed process on how the complaints are recorded and addressed. The main extract from the charter: 5- Complaints Handling Process 5.1- Lodging a Complaint End-user Customers can lodge complaints by calling 800 TASLEEM (8275336), emailing customerservice@tasleem.ae, or visiting the Tasleem website: www.tasleem.ae. Bulk-offtake customers will be able to log complaints by calling 800 Tabreed (8227333), sending an email to customerservice@tabreed.ae, or contacting their designated Account Manager. The complaint can be assigned, tracked, and monitored by the Customer Service team leads. When a call type is selected in the billing system, the operator has the option to list the complaint. All complaint types are included in the complaint reports. 5.2- Receipt and Acknowledgment of Complaint Customers will receive an acknowledgment receipt of their complaint via either a phone call or an e-mail. This acknowledgement is made on the day the complaint is lodged, provided it is a business day, or the following business day if the complaint was made during the weekend or on a public holiday. 5.3- Accessibility to the Complaint

	End-user Customers can request copies of their complaint log by emailing customerservice@tasleem.ae or calling 800 TASLEEM (8275336) during normal business hours. Bulk-offtake Customers can request copies of their complaint log by emailing customerservice@tabreed.ae or calling 800 TABREED (8227333) during normal business hours
Date(s) of stakeholder consultation	19-September-2023
Communication of monitored results	All the communication is recorded in the excel sheet available with the stakeholder team.
Consultation records	All the communication is recorded in the excel sheet available with the stakeholder team.
Stakeholder input	All the inputs were considered and responded accordingly. The stakeholder inputs process is continuous process and the comments will be recorded and appropriately responded.

2.1.3 Free, Prior, and Informed Consent

Consent	Project has received all the required consents from government authorities.
Outcome of FPIC	Project has received all the required consents from government authorities. Project has not encroached on land and relocated people.

2.1.4 Grievance Redress Procedure

Grievances received	Resolution and outcome
No	The project owner has operating complaint procedure. The same is used for the project. No complaint has been received related with the project.

2.1.5 Public Comments

Summary of comments received	Actions taken
No public comments have been received for the project	

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

The PP has been working for last 25 years in similar type of project activities. Tabreed is an ISO certified organisation for environmental management. Its strong management has successfully implemented various similar type of projects.

Tabreed currently delivers over 1.308 million refrigeration tons of cooling, across 90 plants located throughout the region, cooling iconic infrastructure projects such as Sheikh Zayed Grand Mosque, Burj Khalifa, The Dubai Mall, Dubai Opera, Cleveland Clinic, Ferrari World, Yas Mall, Aldar HQ, Etihad Towers, Marina Mall, World Trade Center in Abu Dhabi featuring the Burj Mohammed Bin Rashid, Dubai Metro, Dubai Parks & Resorts, and the Jabal Omar Development in the Holy City of Mecca, alongside several other hotels, hospitals, residential and commercial towers[3].

2.2.2 Risk assessment

	Risk identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	No risk identified	The project is energy efficient district cooling. In absence of the project also the stakeholders would have been using the cooling solutions. Therefore, no risk is identified for wellbeing of stakeholders.
Risks to stakeholder participation	No risk identified	Project is energy efficiency and operation is limited to expert operators and therefore no risk to stakeholder participation.
Working conditions	No risk identified	The project owner is listed public company and abide by all the legal requirements.
Safety of women and girls	No risk identified	The service provided by project is common to all irrespective or gender and therefore no risk identified. Furthermore, the projects instances are in the Abu Dhabi city, which is one of the

[3] https://www.tabreed.ae/

	Risk identified	Mitigation or preventative measure(s) taken
		safest cities in the world and project instances access is secured and therefore no risk for safety of women and girls.
Safety of minority and marginalized groups, including children	No risk identified	The service provided by project is common to all irrespective or gender and therefore no risk identified. Furthermore, the projects instances are in the Abu Dhabi city, which is one of the safest cities in the world and project instances access is secured and therefore no risk for safety of minority and marginalized groups, including children.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	No risk identified	The project is housed in dedicated building and following all the environmental approvals and therefore no risk identified. Furthermore, project activity is reducing the electricity consumption and therefore ha positive impact on the environment.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified[4]	Mitigation or preventative measure(s) taken
Discrimination	In UAE labour laws prohibit any type of discrimination and harassment. Federal Decree-Law No. 33/2021, known as the Labour law, expressly addresses discrimination in various aspects, emphasizing the importance of treating employees fairly and equally. Article 4 of	https://www.tabreed.ae/wp-content/uploads/2024/06/Tabreed-ESG-Report-2023.pdf Page 34 of publicly available ESG report has

[4] The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

	the Labour Code prohibits discrimination based on race, colour, sex, religion, national origin, social origin, or disability. Project owner is listed company and abiding by all the applicable law and therefore no discrimination and sexual harassment occurred and will occur in the future.	provided detailed explanation on Discrimination. No complaints received.
Sexual harassment	The project owner has been implementing similar projects in last 25 years and have sufficient experience on managing all the requirements.	https://www.tabreed.ae/wp-content/uploads/2024/06/Tabreed-ESG-Report-2023.pdf Page 40 of publicly available ESG report has provided detailed explanation on practices of human rights in Tabreed. No complaints received.
Gender equity in labor and work	Tabreed believes in gender equality and publicly publishing the ESG related information. https://www.tabreed.ae/wp-content/uploads/2023/07/Tabreed_ESG_Report_2022.pdf	https://www.tabreed.ae/wp-content/uploads/2024/06/Tabreed-ESG-Report-2023.pdf Page 34 of publicly available ESG report has provided detailed explanation on equity. No complaints received.
Forced labor	In UAE laws prohibit any type of forced labor. The project owner is abiding by all the applicable laws.	https://www.tabreed.ae/wp-content/uploads/2024/06/Tabreed-ESG-Report-2023.pdf Page 40 of publicly available ESG report has provided detailed explanation on practices of human rights in Tabreed. No complaints received.

Child labor	In UAE laws prohibit any type of child labor. The project owner is abiding by all the applicable laws.	https://www.tabreed.ae/wp-content/uploads/2024/06/Tabreed-ESG-Report-2023.pdf Page 40 of publicly available ESG report has provided detailed explanation on practices of human rights in Tabreed. No complaints received.
Human trafficking	In UAE laws prohibit any type of human trafficking. The project owner is abiding by all the applicable laws.	https://www.tabreed.ae/wp-content/uploads/2024/06/Tabreed-ESG-Report-2023.pdf Page 40 of publicly available ESG report has provided detailed explanation on practices of human rights in Tabreed. No complaints received.

2.3.2 Human Rights

Tabreed respects internationally recognised human rights in its relations with public authorities. Tabreed is committed to adhering to international laws and conventions, including, but not limited to, the International Labour Organization (ILO) core conventions: Conventions No. 29 and No. 105 on the Abolition of Forced Labour; Conventions No. 138 and No. 182 on the Abolition of Child Labour; and Conventions No. 100 and No. 111 on Non-discrimination.

More information available on https://www.tabreed.ae/wp-content/uploads/2024/06/Tabreed-ESG-Report-2023.pdf

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	The PP has established practice to manage the human rights related aspects. These are publicly disclosed.

2.3.3 Indigenous Peoples and Cultural Heritage

The project has received all the approvals for the structures of the building from the respective authorities. This approval confirms that all the required regulations have been followed.

Risks identified	Mitigation(s) or preventative measure taken
No risk identified	The PP has established practice to manage the human rights and cultural heritage related aspects. With all the approvals in place the PP has no identified risk.

2.3.4 Property Rights

Project owner has all the rights on the project. The land has been leased and the construction has been carried out by project owner. As the land is allocated and registered in the government system, the approval reflects the ownership on land. For the project instances the project proponent has full ownership of land usage.

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	As the project requires various government approval and it start with the land approval. With the land approval in place, the PP has property rights.

2.3.5 Benefit Sharing

Not applicable.

The project is energy efficiency in district cooling plant, which is owned and operated by the PP. All the investment is made by PP in installation, commissioning and operation.

Summary of the benefit sharing plan	No applicable
Benefit sharing during the monitoring period	Not applicable

2.4 Ecosystem Health

	Risk identified	Mitigation or preventative measure(s) taken during the monitoring period
Impacts on biodiversity and ecosystems	No risk identified	The land used was already developed and no impact on biodiversity due to project.

Soil degradation and soil erosion	No risk identified	The land used was already developed and no impact on biodiversity due to project.
Water consumption and stress	No risk identified	The project will be consuming water. The water will be coming from sea water desalination plant. No impact on potable water.

2.4.1 Rare, Threatened, and Endangered species

The project is not located in, or adjacent to habitats for rare, threatened, or endangered species

Species or habitat	Not applicable
Areas needed for habitat connectivity	The land was already developed.

Risks related to habitats for rare, threatened, and endangered species, and for areas for habitat connectivity

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	Not applicable	Land was already developed.
Areas for habitat connectivity	Not applicable	Land was already developed.

2.4.2 Introduction of species

Not applicable

2.4.3 Ecosystem conversion

Not applicable

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

This monitoring period would be for two instances. Grouped project was registered on 12-July-2024. The project start date of first crediting during this monitoring is 25-May-2023. PA 01 was checked during the time of validation; the second project activity is new inclusion. Both the project instances are included in this monitoring report. Timeline of project activity instances illustrate as the table below:

Project instances	Project instance PA01 (SeaWorld)	Project instance PA02 (Saadiyat)
First crediting period start date	25-May-2023	25-May-2023
Project instances start and inclusion date	25-May-2023	03-May-2024
Capacity	9000 TR	9000 TR

No other changes (e.g., to project proponent or other entities) has happened for the project activity during this monitoring period.

3.2 Deviations

3.2.1 Methodology Deviations

The deviations have been included during the registration of the project. These are included below. No new deviation is proposed.

1. <u>Grid emission factor</u>

The methodology refers to CDM Tool 7 for electricity grid emission factor. However, the data is not available in public domain to calculate the grid emission factor and therefore project owner is not able to calculate this factor.

The project owner request to use the publicly available grid emission factor for the project, which will be fix for first crediting period. For future crediting period project owner anticipates that the data will be available to calculate the grid emission factor and then the Tool 7 or any other Verra guidance will be used for the project.

Based on publicly available information, the following data is available for UAE grid emission factor:

Source	Latest year for which emission factor available	Value
Third update for second NDC of the UAE – 2023 page 17[5]	2019	0.55 tCO_2/MWh
Energy profile, United Arab Emirates published by IRENA – 8th August 2023 page 3[6]	2021	457 tCO_2/GWh (0.457 tCO_2/MWh)

The Project will consume less electricity with respect to the baseline and therefore lower grid emission factor will result in lower baseline emissions and lower emission reductions and therefore the lower grid emission factor will be conservative of therefor lower value (IRENA) is selected. As IRENA value is latest and conservative therefore this has been used for the first crediting period.

2. Benchmarking Boundary

As per the Methodology the default benchmark boundary is a city where the project activity is located which is Abu Dhabi.

No definition is provided related with the city.

The UAE is divided into seven administrative divisions (Abu Dhabi, Dubai, Sharjah, Ajman, Umm Al Quwain, Ras Al Khaimah and Fujairah), referred as emirates. Each emirates have its own rules and regulations and clear demarcation of boundary.

- Therefore, Abu Dhabi Emirate is considered as city for methodological requirement and benchmark boundary. As this is grouped project and therefore the boundary is limited to Abu Dhabi Emirate, United Arab Emirates.

3.2.2 Project Description Deviations

No project description deviation in this monitoring period.

[5] https://unfccc.int/sites/default/files/NDC/2023-07/Third%20Update%20of%20Second%20NDC%20for%20the%20UAE_v15.pdf (page 17)

[6] https://www.irena.org/-/media/Files/IRENA/Agency/Statistics/Statistical_Profiles/Middle-East/United-Arab-Emirates_Middle-East_RE_SP.pdf (page 3)

3.3 Grouped Projects

No.	Eligibility criterion - Category	Eligibility criterion - Required condition	Justification for PA02
1	Applicability Conditions of the applied methodology.	The project activity instances shall meet applicability conditions for applicable methodology (AM 0117 version 02) as defined in section 2.2.	The applicability for the PA 02 is demonstrated in the table below. PA02 meets the eligibility condition.
2	Geographical Area	The project activity instances for the grouped project will be located in the geographic boundary of Abu Dhabi Emirates and owned by the project owner or the subsidiary of the project owner.	The project is located in the Abu Dhabi. The coordinates are included in the Monitoring report. PA02 meets the eligibility condition.
3	Baseline scenario	All Project Activity Instances shall meet the baseline definition as defined in the respective methodology. The baseline scenario is based on technology penetration.	Latest baseline scenario assessment report is available which has been used during the validation and is applicable for the 3 years of monitoring. The baseline scenario is same as in registered PD. PA02 meets the eligibility condition.
4	Technology type:	The project activity instances to be included in the grouped project activity is the implementation of new district cooling plant with or without thermal energy storage.	As per the technical reports and physical installation it is clear that project is new district cooling plant PA02 meets the eligibility condition.
5	Additionality	The project activity instances to be added as part of the grouped project will meet additionality criteria as set out in the methodology. Additionality for this instance has been demonstrated by using requirements provided in para 5.2.1; Step 1:	The project instance 2 (PA02) meets the additionality as per the third-party report for the baseline technology assessment. The technology penetration is less than 20% and the SEER is higher than benchmark SEER.

No.	Eligibility criterion - Category	Eligibility criterion - Required condition	Justification for PA02
		Penetration and Performance assessment, in the applied methodology. a. The seasonal energy efficiency ratio (SEER) of the DCP involved in the project activity is higher than a benchmark SEER and (b) The share of the district cooling technologies at the moment of the project registration is less than 20 per cent of all cooling technologies within the benchmark boundary in terms of cooling output. The future instances shall comply above requirement of methodology for additionality demonstration.	PA02 meets the eligibility condition.
6	Start date	The start date of each project activity instance under the grouped project should not be prior to the start date of the grouped project. The start date of each project activity instance will be determined through documentary evidence.	The PA02 start date is 03-05-2024 as per the commissioning document. PA02 meets the eligibility condition.
7	Capacity Limits	No capacity limits are mentioned in the applied methodology and therefore no capacity limit will be applied for inclusion of new project activity instances.	Not applicable PA02 meets the eligibility condition.
8	Stakeholder consultation:	The local stakeholder consultation will be conducted before inclusion of any new project instance.	Meeting notices, Meeting presentations and photos have been provided. The stakeholder consultation has happened on 19-september-2023. The PP has system of

No.	Eligibility criterion - Category	Eligibility criterion - Required condition	Justification for PA02
			receiving the comments continuously. PA02 meets the eligibility condition. The PP has applied for exemption for this requirement from Verra. Verra approved the exemption for this verification on 8th April 2025. The stakeholder consultation will be conducted before second verification. Therefore, this eligibility condition is justified for this verification period.
9	Environmental Impact:	Every new project instance will follow the host country's environmental regulations related with environmental approvals	The PA02 has received the required approvals as per host country requirements. PA02 meets the eligibility condition.
10	Sampling:	No sampling is used for the emission reduction calculation. All the project instances shall have the monitoring parameters as explained in monitoring section of the methodology AM0117 version 02	Not applicable PA02 meets the eligibility condition.
11	Conditions to avoid double counting - Unique Identification:	All the new project instance will be named uniquely based on location as the PI 1 is called 'Sea World District Cooling Plant'. To prevent double counting of District Cooling Plants under the project, District cooling plant shall be uniquely marked, and new project instance implementers	As per the location of PA02, no carbon project is implemented in the same location. No double counting of the emission reductions from project. PA02 meets the eligibility condition.

No.	Eligibility criterion - Category	Eligibility criterion - Required condition	Justification for PA02
		shall provide assurances/declarations that they are not included in any other CDM or Voluntary GHG offset schemes outside of this project.	
12	Programme management- Approval of new project instance by project proponent; collection of data and record keeping:	The project proponent approves each project instance to be included into its registered group project activity. Project proponent Tabreed will be responsible for keeping all the emission reduction data, including the details of project instances start date, commissioning document and other required information.	Tabreed is managing all the data and approved inclusion of PA02 project instance in the grouped project. PA02 meets the eligibility condition.
13	Conditions to provide an affirmation that funding from Annex I parties, if any, does not result in a diversion of official development assistance.	under this project, new project instance operators shall be required to provide an affirmation that funding from annex I parties, if any, does not result in a diversion of official development assistance	The PP has provided an affirmation that funding from annex 1 parties has not been used for PA02. PA02 meets the eligibility condition.
14	VCU ownership	End users receiving Cooling demand under the specific new project instance contractually cede their rights to claim and own emission reductions under the VCS to the project proponent of the group project.	The PP has already signed the agreement for supply of services to the new customers. VCU ownership is with PP. PA02 meets the eligibility condition.
15	Awareness and agreement of those operating a new project instance on group project subscription	Contractual provisions to ensure that those operating the new project instance are aware and have agreed that their activity is being subscribed to the new project instance.	The operator is PP for PA02. This is not applicable for PA02. PA02 meets the eligibility condition.

No.	Eligibility criterion - Category	Eligibility criterion - Required condition	Justification for PA02

Applicability of Methodology

Methodology ID	Applicability condition	Justification of compliance
AM0117	The methodology is applicable to project activities that reduce CO_2 emissions by means of one, or a combination, of the following measures: (a) Introduction of new district cooling system(s) that supply cooling to residential and commercial consumers through a new dedicated distribution network; (b) Introduction of new district cooling system(s) that supply cooling to residential and commercial consumers through an existing dedicated distribution network; (c) Expansion of the existing district cooling system(s) by adding a new district cooling plant(s) with or without expanding a dedicated distribution network.	Applicable The Project is new district cooling system that supply cooling to residential and commercial consumers through dedicated distribution network.
AM0117	Emission reductions that are gained due to the switch of the energy sources shall not be claimed by applying this methodology alone. Therefore, emission reductions due to displacement of the baseline power	Not applicable The electricity from the grid is the power source in the baseline scenario as well as in the project instance and

Methodology ID	Applicability condition	Justification of compliance
	source can be claimed by the means of the application of this methodology in combination with another relevant approved methodology. In doing so, interactive effects shall be considered as per the "Guidelines for the consideration of interactive effects for the application of multiple CDM methodologies for a programme of activities". For example, if the project district cooling plant is powered partially or completely by a dedicated renewable energy power plant, a project proponent may wish to consider small scale methodology 'AMS-I.F: Renewable electricity generation for captive use and mini-grid' to account for emission reductions due to shift of the power source. Otherwise, the most conservative electricity emission factor should be applied to determine emission reductions.	therefore project will not replace the energy source and therefore this condition is not applicable.

3.4 Baseline Reassessment

Did the project undergo baseline reassessment during the monitoring period?

☐ Yes ☒ No

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	Building Type
Data unit	-
Description	The building types is categorized as new. The project will be providing the cooling to new buildings.

Source of data	District cooling chilled water piping network provided by project owner Tabreed
Value applied	-
Justification of choice of data or description of measurement methods and procedures applied	The data is provided by the project owner and checked through the cooling service bills.
Purpose of data	• Calculation of baseline emissions
Comments	Data will be stored and maintained during verification.

Data / Parameter	Baseline cooling technologies
Data unit	-
Description	The data collection has been done by third party based on various data sources, including Abu Dhabi Department of Energy (DoE), Abu Dhabi Industrial Strategy (ADIS), Abu Dhabi Department of Economic Development (ADDED), Abu Dhabi Industrial Development Bureau (IDB) and Global Market Insights (third party consultant). The data from the third party is provided in consolidated form as per the table below due to maintain confidentially of specific building. The data is segregated based on the baseline cooling technology and type of building. The data set includes from 2019. The independent third party has grouped the baseline cooling technologies and cooling output within Abu Dhabi.
Source of data	Third Party Survey Report
Value applied	(see table below)
Justification of choice of data or description	Data is provided by third party independent company.

Technology	SEER	Cooling Output in million TR	Cumulative Cooling Output (million TR)
District Cooling	4.84	0.63	0.63
Split Units	4.51	0.58	1.21
Centralized Water Cooled	4.31	0.27	1.48
Others (Ducted & Packaged Cooling Systems)	3.52	1.03	2.51
Air Cooled Stand Alone	2.26	1.04	3.55
Total		3.55	

of measurement methods and procedures applied	
Purpose of data	• Calculation of baseline emissions
Comments	Data will be stored and maintained during verification.

Data / Parameter	$EF_{grid,CM,y}$
Data unit	t CO_2/MWh
Description	Combined margin emission factor for the grid
Source of data	IRENA UAE Energy profile published 8th August 2023
Value applied	0.457
Justification of choice of data or description of measurement methods and procedures applied	Conservative publicly available information
Purpose of data	• Calculation of baseline and project emissions
Comments	This will be fixed for first crediting period.

Data / Parameter	Q_{water}
Data unit	Ton
Description	Maximum designed quantity of freshwater to be used in the project system
Source of data	PA01: Concept Design Report Page 28 for 9000 TR PA02: Design Report page 4 for 9000 TR project activity (Table 3)
Value applied	PA01: 1620.52 PA02: 2260
Justification of choice of data or description of measurement methods and procedures applied	Design data

Purpose of data	• Calculation of leakage
Comments	This data will be used for first year only as per the methodology.

4.2 Data and Parameters Monitored

Data / Parameter	OPC,r,y
Data unit	TRh
Description	Cooling output of new district cooling plant r in year y
Source of data	Data is based on cooling output meters recorded in AD026 and AD021-.
Description of measurement methods and procedures to be applied	Both the project instances have already implemented SAP system which takes the data from the meters and prepare the daily reports. Similarly, SAP system records the data on consumers as well. The conservative (minimum between production and consumption) values are used in the emission reduction calculation as per QA/QC procedure.
Frequency of monitoring /recording	Continuous monitoring and daily recording
Value applied	<table><tr><td>PA</td><td>25-May-2023 to 31-December-2023 ($MWh_{cooling}$)</td><td>01-January-2024 to 31-July-2024 ($MWh_{cooling}$)</td></tr><tr><td>PA01</td><td>115,566</td><td>93,244</td></tr><tr><td>PA 02</td><td>0</td><td>37,370</td></tr></table> Values after delayed calibration adjustment <table><tr><td>PA</td><td>25-May-2023 to 31-December-2023 ($MWh_{cooling}$)</td><td>01-January-2024 to 31-July-2024 ($MWh_{cooling}$)</td></tr><tr><td>PA01</td><td>115,335</td><td>93,057</td></tr><tr><td>PA 02</td><td>0</td><td>37,295</td></tr></table>
Monitoring equipment	Cooling Output Meters
QA/QC procedures	Meter Serial number: PA01

to be applied

Meter ID	Description	Unit of measurement	Make	Model	Accuracy class	Calibration date	Next Calibration date
NTP01BT1	Sea World Plant Main BTU Meter	Ton Hours (TRh)	Endress+Hauser	EngyCal RH33	0.2	June 4, 2024 Delayed Calibration adjustment applied.	June 4, 2026
CUC01BT0008000317	Customer - Sea World Plant	Ton Hours (TRh)	Endress+Hauser	EngyCal RH33	0.2	June 4, 2024 Delayed Calibration adjustment applied	June 4, 2026

PA02

Meter ID	Description	Unit of measurement	Make	Model	Accuracy Class	Calibration date	Next calibration date
NTP01BT1	AD-021 Plant Main BTU Meter	Ton Hours (TRh)	Endress +Hauser	EngyCal RH33	0.2	OEM certified - 03/07/2023	- 03/07/2025
CUC02BT0008000228	Customer - SB10	Ton Hours (TRh)	Kamstrup	Multical 801	0.2	13-07-2023	13-07-2025
CUC01BT0008000234	Customer - SB19	Ton Hours (TRh)	Kamstrup	Multical 801	0.2	13-07-2023	13-07-2025
CUC11BT0008000227	Customer - SB21	Ton Hours (TRh)	Kamstrup	Multical 801	0.2	13-07-2023	13-07-2025

CUC14BT0008000235	Customer - C11	Ton Hours (TRh)	Kamstrup	Multical 803	0.2	May 7, 2024	May 7, 2026
CUC17BT0008000235	Customer - C01	Ton Hours (TRh)	Kamstrup	Multical 803	0.2	April 7, 2024	April 7, 2026
CUC01BT0008000519	Customer - RBH	Ton Hours (TRh)	Kamstrup	Multical 803	0.2	OEM certified - New – June 6, 2024	June 6, 2026

Cross check procedure: The cooling output meters are installed at Energy Transfer Stations. The monthly data will be consolidated from all the energy transfer stations (ETS) connected with the district cooling plant. The lower value between consolidated ETS data and District cooling plant will be considered (conservative). This will be crosschecked with invoice as well which will be issued to the user of cooling. Most conservative value will be used for emission reduction.

Calibration frequency: As per Department of energy 'the metering and data exchange code' dated dec 2021 (page 110), the onsite check frequency is 2 years and full calibration frequency is 5 years. This is followed for the project activity

Purpose of data	• Calculation of baseline emissions
Calculation method	As per equation in Baseline Emission section above. The TRh value will be converted to MWh by multiplication of conversion factor 0.003517
Comments	-

Data / Parameter	$EC_{PJ,j,y}$
Data unit	MWh
Description	Quantity of electricity consumed by the new district cooling plant j in year y
Source of data	Monthly ADDC Invoices
Description of measurement methods	This will be monitored and calculated (addition of the electricity data as two meters are available) through online monitoring system.

and procedures to be applied	
Frequency of monitoring/recording	Continuous monitoring and daily recording for Internal meters. Monthly for ADDC meter.
Value applied	<table><tr><td>PA</td><td>25-May-2023 to 31-December-2023 (MWh)</td><td>01-January-2024 to 31-July -2024 (MWh)</td></tr><tr><td>PA01</td><td>28,927</td><td>22,070</td></tr><tr><td>PA 02</td><td>0</td><td>11,892</td></tr></table>
Monitoring equipment	Electricity Meters
QA/QC procedures to be applied	Meter number: PA01 <table><tr><td>Meter ID (AD026)</td><td>Description</td><td>Unit of measurement</td><td>Make</td><td>Model</td><td>Accuracy class</td><td>Calibration date</td><td>Next calibration date</td></tr><tr><td>EIM0001K</td><td>Plant - ADDC Electric Meter Incomer-1 (internal SAP meter)</td><td>kWh</td><td>CEWE D1189857</td><td>ELITE 440</td><td>0.2</td><td>07.09.2022</td><td>07.09.2024</td></tr><tr><td>EIM0002K</td><td>Plant - ADDC Electric Meter Incomer-2 (internal SAP meter)</td><td>kWh</td><td>CEWE D1189855</td><td>ELITE 440</td><td>0.2</td><td>07.09.2022</td><td>07.09.2024</td></tr><tr><td>E2T12D000237</td><td>Plant - ADDC Electric Meter Incomer-1</td><td>Kilowatt Hour</td><td>Elster</td><td>A1800</td><td>According to IEC 62053-21 &</td><td>ADDC custody</td><td></td></tr></table>

					62053-22		
E2T12D000238	Plant - ADDC Electric Meter Incomer-2	Kilowatt Hour	Elster	A1800	According to IEC 62053-21 & 62053-22	ADDC custody	

PA02

Meter ID (AD021)	Description	Unit of measurement	Make	Model	Accuracy class	Calibration date	Next calibration date
EIM0001K	Plant - ADDC Electric Meter Incomer-1	kWh	Schneider MZ-2302A044-01	10N92030	0.2	20.02.2024	20.02.2026
EIM0002K	Plant - ADDC Electric Meter Incomer-2	kWh	Schneider MZ-2302A042-01	10N92030	0.2	20.02.2024	20.02.2026
EIM0003K	Plant - ADDC Electric Meter Incomer-3	kWh	Schneider MZ-2302A053-01	10N92030	0.2	05.03.2024	05.03.2026
EIM0005K	Plant - ADDC Electric Meter Incomer-Booster Pump	kWh					

	Station							
E2T12D000153	ADDC Meter 1	kWh	Elster	A1800	According to IEC 62053-21 & 62053-22	ADDC custody	
E2T12D000154	ADDC Meter 2	kWh	Elster	A1800	According to IEC 62053-21 & 62053-22	ADDC custody	
E2T12D000155	ADDC meter 3	kWh	Elster	A1800	According to IEC 62053-21 & 62053-22	ADDC custody	
I1C14D005987	ADDC booster pump	kWh	Iskra	MT 372	according to IEC62052-11	ADDC custody	

Calibration: Calibration frequency: As per Department of energy 'the metering and data exchange code' dated dec 2021 (page 110), the onsite check frequency is 2 years and full calibration frequency is 5 years. This is followed for the project activity

Cross check procedure: The power monitoring units are installed parallel to ADDC meters at power incomers. The values are recorded daily at the site office. This monthly value will be crosschecked with ADDC invoice reading. In case of any differences higher value (conservative) will be considered in emission reduction calculation.

Purpose of data	• Calculation of project emissions

Calculation method	KWh value will be converted in MWh by dividing by 1000.
Comments	-

Data / Parameter	$TDL_{B,y}$ and TDLy
Data unit	%
Description	Average technical transmission and distribution losses for providing electricity in year y
Source of data	Annual average value based on the most recent data available within the host country;
Description of measurement methods and procedures to be applied	This will be based on published reports.
Frequency of monitoring/recording	Annually. In the absence of data from the relevant year, most recent figures should be used, but not older than 5 years
Value applied	Transmission loss: 2.04% Distribution loss: 7.2%
Monitoring equipment	- Not applicable
QA/QC procedures to be applied	Using credible data for electricity system (default value)
Purpose of data	• Calculation of baseline and project emissions
Calculation method	Transmission loss and distribution loss values will be added and used in the equation.
Comments	Data from Department of Energy's technical report 2023 is used which is the latest available regulatory information on transmission and distribution.

Data / Parameter	$SEER_{B,i}$
Data unit	MWh/MWh
Description	Benchmark Seasonal Energy Efficiency Ratio of the baseline cooling technology i
Source of data	From manufacturers of the baseline cooling technology

Description of measurement methods and procedures to be applied	The value will be provided by independent entity.
Frequency of monitoring/recording	Every 3 years
Value applied	2.26
Monitoring equipment	Not applicable. Based on third party report.
QA/QC procedures to be applied	Based on third party report
Purpose of data	• Calculation of baseline emissions
Calculation method	*Units will be converted as per the report provided. Based on the present report BTU/W.h is converted to MWh/MWh by dividing it to 3.412.*
Comments	This will be valid for 3 years baseline emission calculations.

Data / Parameter	$R_{k,y}$
Data unit	T
Description	Quantity of refrigerant k used in the project in year y
Source of data	Records from the plant operator
Description of measurement methods and procedures to be applied	Metering will rely on the simple counting of cylinders
Frequency of monitoring/recording	Continuous, As and when cylinder will be used
Value applied	<table><tr><td>PA</td><td>25-May-2023 to 31-December-2023 (MWh)</td><td>01-January-2024 to 31-July-2024 (MWh)</td></tr><tr><td>PA01</td><td>0</td><td>0</td></tr><tr><td>PA 02</td><td>0</td><td>0</td></tr></table>
Monitoring equipment	Supplier
QA/QC procedures to be applied	Crosschecked with purchase records. All meters and scales will be calibrated as per manufacturers' recommendations

Purpose of data	• Calculation of leakage emissions
Calculation method	*The total quantity of refrigerant used to fill the chillers in the project activity will be added.*
Comments	-

Data / Parameter	GWPk,
Data unit	unitless (its ratio)
Description	Global Warming Potential of the refrigerant k (R134A)
Source of data	Records from the plant operator
Description of measurement methods and procedures to be applied	Intergovernmental Panel of Climate Change (IPCC)'s latest reports or any other relevant scientific body's assessment
Frequency of monitoring/recording	As and when new IPCC report published with updated global warming potential values.
Value applied	1530
Monitoring equipment	Not applicable
QA/QC procedures to be applied	This is from authentic sources/default value
Purpose of data	• Calculation of leakage emissions
Calculation method	Not applicable
Comments	AR6 report value for HFC 134a. Shall be updated according to any future COP/MOP decisions

Data / Parameter	EF_{water}
Data unit	tCO_{2e}/t
Description	Emission factor associated with the production of fresh water
Source of data	Emirates water and electricity company
Description of measurement methods	this data is from official source.

and procedures to be applied	
Frequency of monitoring/recording	First monitoring period
Value applied	0.0115
Monitoring equipment	Not applicable. Data is provided by third party.
QA/QC procedures to be applied	This is data from the water producer which is supplier to full Abu Dhabi and will follow its internal QA/QC process
Purpose of data	• Calculation of leakage emissions
Calculation method	Not Applicable
Comments	The publicly available value is used during validation.

4.3 Monitoring Plan

The project is operated by Tabreed which ensures the overall site management in accordance with the UAE Laws and technology providers' guidelines and will perform the monitoring in house.

The project monitoring is in compliance with the monitoring methodology AM0117 Version 2.0. Therefore, all relevant parameters stated in section above are reliably monitored and cross checked through calibrated measurement equipment.

Additionally, all workers will receive the appropriate training for the operation and maintenance of all equipment based on the monitoring equipment maintenance/calibration procedures, the emergency strategy and sampling protocols for waste composition analysis.

All data collected as part of monitoring will be archived electronically and be kept at least for 2 years after the end of the last crediting period.

The monitoring details are below:

1. The requirement of monitoring plan

According to the approved methodology and registered project design, the project participants are monitor the emission reductions (ERs) by methods, indicators, and frequency to ensure project ERs are measurable and real. The monitoring methodology is based on direct measurement of the amount of cooling produced from district cooling system and amount of electricity consumed.

2. Responsibilities of operational and management structure

The project participant has implemented the registered monitoring plan. The plan could be revised according to suggestions from Designated Operational Entity (DOE) and the practical circumstances, in order to keep it consistent, transparent and conservative during the monitoring process.

(1) Principal of the monitoring procedure

The chief operating officer (COO) is the project leader who sets out the responsibility of everyone in the monitoring system and establishes the related documents.

COO ensures that staffs in the monitoring system has the ability to deal with the assigned tasks.

The respective project instances operations supervisor is responsible for aggregating the monitored data daily, monthly, and yearly, archiving and keeping data during the crediting period and two years after. All the data is stored in SAP system.

Operators will be in charge of data supervision, checking and inspecting the system. If necessary, they will have the responsibility for executing the emergency plan and drafting emergency reports.

(2) Executive person of the monitoring procedure

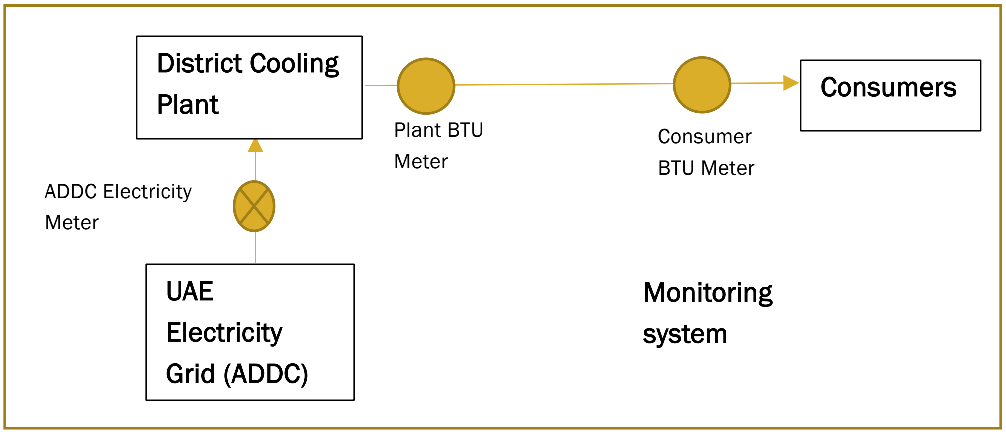
Plant Manager/operation supervisor: A Plant Manager is appointed who is specifically responsible for training, checking the daily operation, reporting forms and archiving emergency situation reports. The Plant Manager reports to the COO about the project performance and monitored data. In the event that non-conformance in the performance to the mentioned procedures and/or functioning problems of the monitoring equipment are identified, the Plant manager will inform the COO about the situation and work out relevant measures to be taken. The Plant manager will also be responsible for aggregating the monitored data monthly and yearly, archiving and keeping data during the crediting period and two years after.

(3) Operators of the monitoring procedure

Operators will take turns to work in the control center 24 hours a day, 7 days a week. They will be in charge of data supervision, filling operation report forms and, checking and inspecting the system. If necessary, they will have the responsibility for executing the emergency plan and drafting emergency situation reports.

3. Monitoring system:

The meter points are mentioned below diagram.



4. **Data collection procedures**

The data are fully recorded and archived automatically and shown in the control system. The data are recorded once per day. SAP system stores all the data and has backup of the data. All data are kept 2 years after the end of the crediting period.

5. **QA/QC**

In order to ensure monitoring plan with high quality, QA/QC measures are carried out in monitoring data recording and checking, equipment calibrating and staff training.

All the monitoring devices (BTU meters and electricity meter) are revenue grade meters used for invoicing and these are calibrated regularly for accuracy according to the national regulations.

Data recording: all data collected are recorded in electronic files which are regularly backed up.

The data are checked by the specific staff every month.

Equipment calibration and maintenance: BTU meters and other critical project equipment are subject to regular maintenance and testing according to technical specifications from the manufactures to ensure accuracy and good performance.

To assist in future verification, all the data files of project monitoring will be kept in the archives and checked by Plant Manager.

6. **Data Management**

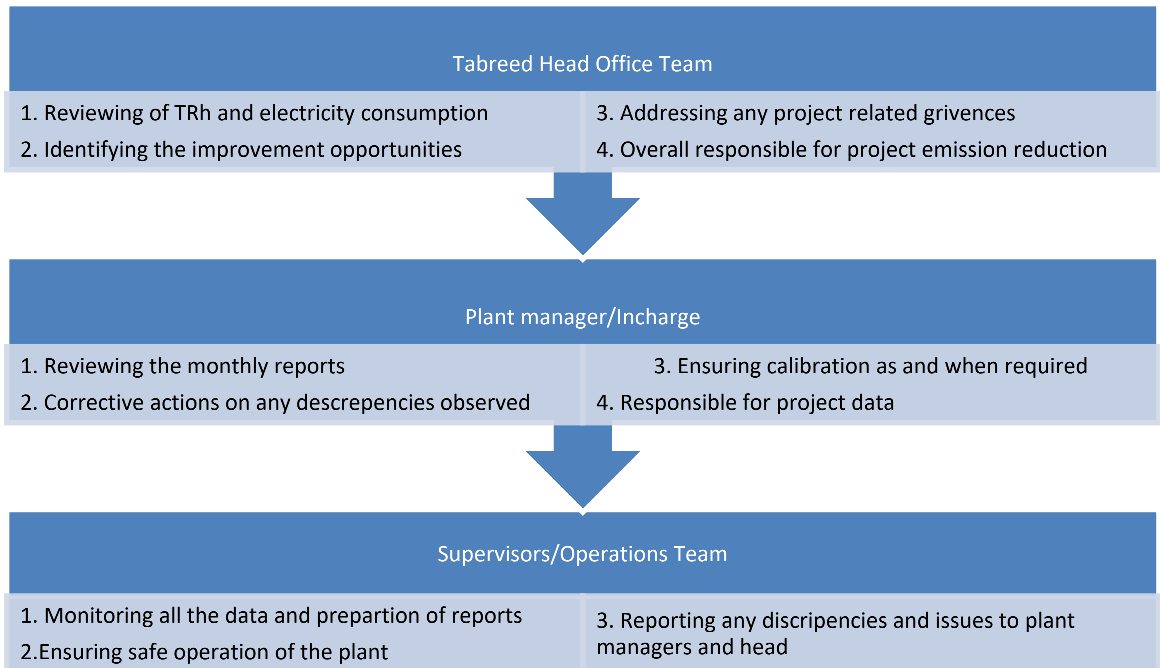
All the data is stored in SAP system. Dedicated staff are working for overall management of SAP data system and keeping all monitored data collected as part of monitoring archived electronically and be kept at least for 2 years after the end of the last crediting period.

Electronic data and documents will be regularly copied and archived and kept at least two years after the end of the last crediting period.

All written data and documents, including electricity receipts for cross-checking, will be copied and archived and kept at least two years after the end of the last crediting period.

7. **Procedure in case of failure**

In the case of a meter in fault, it shall be immediately repaired or replaced with another calibrated meter by a professional engineer, and the cooling produced during the period of erroneous measurement and replacement of the fault meter shall not be accounted for conservative consideration.



5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

. The emission reduction calculation sheet for full monitoring period is submitted.

Baseline emissions are estimated using Approach 1 as the benchmark installation is electricity driven technology.

Total baseline emissions are calculated as follows, using Equation (2) from the methodological tool 05; Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation version 3.0. When applying the tool, parameter QB,y shall correspond to parameter $EC_{BL,k,y,}$ whereas BE_y shall correspond to $BE_{EC,y}$.

$$BE_y = Q_{B,y} \times EF_{B,y} x\ (1 + TDL_{B,y})$$ Equation (1)

Where

BE_y	=	Baseline emissions in year y (tCO_2e/yr)
$Q_{B,y}$	=	Estimated electricity consumption of isolated and less efficient air-cooled reciprocating chiller systems in year y (MWh/yr)
TDL_{BY}	=	Average technical transmission and distribution losses for providing electricity to the baseline in year y
$EF_{EL,y}$	=	Grid CO_2 emission factor calculated in accordance with the “Tool to calculate the emission factor for an electricity system” (Version 7.0) (tCO_2e/MWh)

PA01

Where:		**25/05/2023-31/12/2023**	**01/01/2024-31/07/2024**	**Units**	**SOURCE**
BE_y	Baseline emissions in year *y*	25,477	20,556	tCO2e	Calculated
$Q_{B,y}$	Estimated electricity consumption of isolated and less efficient air-cooled reciprocating chiller systems in year *y*	51,033	41,176	MWh	Calculated based on the equation below
$EF_{B,y}$	Grid CO2 emission factor	0.457	0.457	tCO2e/MWh	Energy profile, United Arab Emirates published by IRENA – 8th August 2023 page 3
$TDL_{B,y}$	Average technical transmission and distribution losses for providing electricity in baseline in year *y*	9.24%	9.24%	%	Department of energy '2023 Annual Technical Report. Page 28 for transmission losses (2.04%) and page 38 for distribution losses (7.2%)

PA 02

Where:		**25/05/2023 - 31/12/2023**	**01/01/2024 -31/07/2024**	**Units**	**SOURCE**
BE_y	Baseline emissions in year *y*	0	8,238	tCO2e	Calculated

Where:		25/05/2023 - 31/12/2023	01/01/2024 -31/07/2024	Units	SOURCE
$Q_{B,y}$	Estimated electricity consumption of isolated and less efficient air-cooled reciprocating chiller systems in year *y*	0	16,502	MWh	Calculated based on the equation below
$EF_{B,y}$	Grid CO2 emission factor	0.457	0.457	tCO2e/MWh	Energy profile, United Arab Emirates published by IRENA – 8th August 2023 page 3
$TDL_{B,y}$	Average technical transmission and distribution losses for providing electricity in baseline in year *y*	9.24%	9.24%	%	Department of energy '2023 Annual Technical Report. Page 28 for transmission losses (2.04%) and page 38 for distribution losses (7.2%)

The estimate cooling output of isolated and less efficiency cooling technologies is determined as follows:

$$Q_{B,y} = \sum_{r} \frac{OPC_{r,y}}{SEER_{B,i}} \qquad \text{Equation (2)}$$

Where:

$Q_{B,y}$ = Energy consumed in baseline by baseline cooling technologies in year y (MWh)

$OPC_{,r,y}$ = Cooling output of new district cooling plant r in year y (MWh)[7]

$SEER_{B,i}$ = Benchmark Seasonal Energy Efficiency Ratio of the baseline cooling technology i, defined as energy output divided by energy input (MWh/MWh)

PA 01

Where:		25-May-2023 to 31-December-2023	01-January-2024 to 31-July-2024	Units	Source

[7] 1 TR= 3.517 kW = 0.003517 MW

$Q_{B,y}$	Energy consumed in baseline by baseline cooling technologies in year y (MWh)	51,033	41,176	MWh	Calculated
$OPC_{r,y}$	Cooling output of new district cooling plant r in year y (MWh)	115,335	93,057	MWh	Measured data as per monitoring plan
$SEER_B$	Benchmark Seasonal Energy Efficiency Ratio of the baseline cooling technology i, defined as energy output divided by energy input (MWh/MWh)	2.26	2.26	MWh/MWh	As per the methodology procedure based on data provided by Third party (Fixed for 3 years)

PA02

Where		25-May-2023 to 31-December-2023	01-January-2024 to 31-July-2024	Units	Source
$Q_{B,y}$	Energy consumed in baseline by baseline cooling technologies in year y (MWh)	0	16,502	MWh	Calculated
$OPC_{r,y}$	Cooling output of new district cooling plant r in year y (MWh)	0	37,295	MWh	Measured data as per monitoring plan
$SEER_B$	Benchmark Seasonal Energy Efficiency Ratio of the baseline cooling technology i, defined as energy output divided by energy input (MWh/MWh)	2.26	2.26	MWh/MWh	As per the methodology procedure based on data provided by Third party (Fixed for 3 years)

The grid CO2 emission factor for UAE grid 0.457 tCO2/MWh is used from International Renewable Energy Agency (IRENA) UAE Energy Profile for 2021. This will be fixed for first crediting period.

As per paragraph 39 of the methodology 'The cooling output of the new district cooling plant can be obtained as follows:
Option 1: Direct measurement of the cooling energy (e.g. MWh or GJ), or
Option 2: Calculated based on measurements of temperature differences, flow rate of chilled water and system operating hours per year:

The project instances are using direct measurement of the cooling energy (option1).

5.2 Project Emissions

Project emissions include emissions from electricity consumption associated with the generation of cooling output in the new district cooling plant.

These emissions are calculated using the "Methodological tool: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" version 3.0.

$$PE_y = PE_{EC,y} + PE_{FC,j,y} \qquad \text{Equation (3)}$$

Where:

PE_y = Project emissions in year y (tCO_2e)

$PE_{EC,y}$ = Emissions from electricity consumption associated with the generation of cooling output in the new district cooling plant(s) (tCO_2e)

$PE_{FC,j,y}$ = CO2 emissions from fossil fuel combustion associated with the generation of cooling output in the new district cooling plant(s) (tCO_2e)

PA01

Where:		**25-May-2023 to 31-December-2023**	**01-January-2024 to 31-July-2024**	**Units**
PE_y	Project emissions in year *y* (tCO_2e)	14,442	11,019	tCO2/year
$PE_{EC,y}$	Emissions from electricity consumption associated with the generation of cooling output in the new district cooling plant(s) (tCO_2e)	14,442	11,019	tCO2/year
$PE_{FC,j,y}$	CO_2 emissions from fossil fuel combustion associated with the generation of cooling output in the new district cooling plant(s) (tCO_2e)	0	0	tCO2/year

PA02

Where:	**Monitoring Period**	**25-May-2023 to 31-December-2023**	**01-January-2024 to 31-July-2024**	**Units**
PE_y	Project emissions in year *y* (tCO_2e)	0	5,937	tCO2/year
$PE_{EC,y}$	Emissions from electricity consumption associated with the generation of cooling output in the new district cooling plant(s) (tCO_2e)	0	5,937	tCO2/year
$PE_{FC,j,y}$	CO_2 emissions from fossil fuel combustion associated with the generation of cooling output in the new district cooling plant(s) (tCO_2e)	0	0	tCO2/year

Emissions from electricity consumption associated with the generation of cooling output in the new district cooling plant(s)

$$PE_{EC,j,y} = EC_{PJ,j,y} \times EF_{EL,y} x \ (1 + TDL_y) \qquad \text{Equation (4)}$$

Where

$EC_{PJ,j,y}$	=	Quantity of electricity consumed by the new district cooling plant j in year y (MWh/yr)
$EF_{EL,y}$	=	Grid CO2 emission factor calculated in accordance with the "Tool to calculate the emission factor for an electricity system" (Version 7.0) (tCO_2e/MWh)
TDL_Y	=	Average technical transmission and distribution losses for providing electricity in year y

PA01

Where		**25-May-2023 to 31-December-2023**	**01-January-2024 to 31-July-2024**	**Units**	**SOURCE**
$PE_{EC,Y}$	Project emissions from electricity consumption associated with the generation of cooling output in the district cooling plant in year y	14,442	11,019	tCO2e	Calculated
$EC_{PJ,j,y}$	Quantity of electricity consumed by the new district cooling plant j in year y	28,927	22,070	MWh	Measured electricity consumption
$EF_{EL,y}$	Grid CO2 emission factor	0.457	0.457	tCO2/MWh	Energy profile, United Arab Emirates published by IRENA – 8th August 2023 page 3
TDL_y	Average technical transmission and distribution losses in year y	9.24%	9.24%	%	Department of energy '2023 Annual Technical Report. Page 28 for transmission losses (2.04%) and page 38 for distribution losses (7.2%)

PA02

Where		**25-May-2023 to 31-December-2023**	**01-January-2024 to31-July-2024**	**Units**	**SOURCE**
$PE_{EC,Y}$	Project emissions from electricity consumption associated with the generation of cooling output in the district cooling plant in year y	0	5,937	tCO2e	Calculated

$EC_{PJ,j,y}$	Quantity of electricity consumed by the new district cooling plant j in year y	0	11,892	MWh	Measured electricity consumption
$EF_{EL,y}$	Grid CO2 emission factor	0.457	0.457	tCO2/MWh	Energy profile, United Arab Emirates published by IRENA – 8th August 2023 page 3
TDL_y	Average technical transmission and distribution losses in year y	9.24%	9.24%	%	Department of energy '2023 Annual Technical Report. Page 28 for transmission losses (2.04%) and page 38 for distribution losses (7.2%)

CO2 emissions from fossil fuel combustion associated with the generation of cooling output in the new district cooling plant(s)

$$PE_{FC,j,y} = \sum FC_{i,j,y} \times COEF_{i,y}$$

$PE_{FC,j,y}$	CO_2 emissions from fossil fuel combustion associated with the generation of cooling output in the new district cooling plant(s) (tCO_2e)
$FC_{i,j,y}$	Is the quantity of fuel type i combusted in process j during the year y (mass or volume unit/yr) C
$COEF_{i,y}$	Is the CO_2 emission coefficient of fuel type i in year y (tCO_2/mass or volume unit)
i	Are the fuel types combusted in process j during the year y

The project activity has no fossil fuel consumption[8] during the monitoring period and will be using only electricity therefore $PE_{FC,j,y}$= o

[8] As the project is designed for operation by directly electricity and therefore this parameter will not be monitored during the monitoring period.

5.3 Leakage Emissions

Leakage emissions are calculated as follows:

$$LE_y = LE_{Ref,y} + LE_{Ref,sc} + LE_{water}$$ Equation (5)

Where:

LE_y = Leakage emissions in year y (tCO_{2e})

$LE_{Ref,y}$ = Refrigerant leakage emissions from the project in year y (tCO_{2e})

$LE_{Ref,sc}$ = Refrigerant leakage emissions from the baseline cooling equipment that is scrapped as a result of the project activity (only accounted in the first year of the crediting period) (tCO_{2e})

$LE_{water,y}$ = Emissions due to the freshwater usage in the project system in year y (tCO_2e)

PA01

Where:			**25-May-2023 to 31-December-2023**	**01-January-2024 to 31-July-2024**	**Units**
LE_y	=	Leakage emissions in year *y* (tCO_2e)	**19**	**0**	tCO2
$LE_{Ref,y}$	=	Refrigerant leakage emissions from the project in year *y* (tCO_2e)	0	0	tCO2
$LE_{Ref,sc}$	=	Refrigerant leakage emissions from the baseline cooling equipment that is scrapped as a result of the project activity (only accounted in the first year of the crediting period) (tCO_2e)	0	0	tCO2
$LE_{water,y}$	=	Emissions due to the freshwater usage in the project system in year *y* (tCO_2e)	19	0	tCO2

PA02

Where:			**25-May-2023 to 31-December-2023**	**01-January-2024 to 31-July-2024**	**Units**
LE_y	=	Leakage emissions in year *y* (tCO_2e)	**0**	**26**	tCO2
$LE_{Ref,y}$	=	Refrigerant leakage emissions from the project in year *y* (tCO_2e)	0	0	tCO2
$LE_{Ref,sc}$	=	Refrigerant leakage emissions from the baseline cooling equipment that is scrapped as a result of the project activity (only	0	0	tCO2

		accounted in the first year of the crediting period) (tCO_2e)			
$LE_{water,y}$	=	Emissions due to the freshwater usage in the project system in year *y* (tCO_2e)	0	26	tCO2

Emissions due to the refrigerants leakage during the project activity is calculated as:

$$LE_{Ref,y} = \sum_{k} R_{k,y} \times GWP_k \quad \text{Equation (6)}$$

Where:

$LE_{Ref,y}$ = Refrigerant leakage emissions from the project in year y (tCO2)

$R_{k,y}$ = Quantity of refrigerant k filled in the district cooling system year y (t)

GWP_k = Global Warming Potential of the refrigerant k

PA01

Monitoring Period	Units	**25-May-2023 to 31-December-2023**	**01-January-2024 to 31-July-2024**	
Type of refrigerant		R 134 A	R 134 A	Actual Data from Chiller Manufacturer's
Global warming potential ($GWP_{k,y}$)	tCO2/ ton of HFC 134 a	1530	1530	AR 6 Report https://ghgprotocol.org/sites/default/files/2024-08/Global-Warming-Potential-Values%20%28August%202024%29.pdf
Leakage of Refrigerant (charged quantity)	kg	0.00	0.00	Monitored data
	tons	0.000	0.000	
Emissions ($LE_{ref,y}$)	tCO2	0	0	

PA02

Monitoring Period	Units	**25-May-2023 to 31-December-2023**	**01-January-2024 to 31-July-2024**	

Type of refrigerant		R 134 A	R 134 A	Actual Data from Chiller Manufacturer's
Global warming potential ($GWP_{k,y}$)	tCO2/ ton of HFC 134 a	1530	1530	AR 6 Report https://ghgprotocol.org/sites/default/files/2024-08/Global-Warming-Potential-Values%20%28August%202024%29.pdf
Leakage of Refrigerant (charged quantity)	kg	0.00	0.00	Monitored data
	tons	0.000	0.000	
Emissions ($LE_{ref,y}$)	tCO2	0	0	

Emissions due to the refrigerant leakage from the baseline cooling equipment that is scrapped as a result of the project activity9 are calculated (only for the first year of the crediting period) as below:

$$LE_{Ref,SC} = \sum_{z} R_z \times GWP_z$$ Equation (7)

Where:

$LE_{Ref,SC}$ = Refrigerant leakage emissions from the baseline individual cooling equipment that is scrapped as a result of the project activity (tCO2e)

R_z = Quantity of refrigerant z leaked from the baseline individual cooling equipment that is scrapped as a result of the project activity (t)

GWP_z = Global Warming Potential of the refrigerant z

This is not applicable for project as no baseline equipment are scrapped for project instances PA01 and PA02. Therefore:

$$LE_{Ref,SC} = 0 \text{ tCO2/year}$$ Equation (8)

Emissions due to use of freshwater shall be considered in sites where freshwater is produced through the desalination. As emissions associated with the production of make-up water are considered

[9] Emissions due to the possible refrigerant leakage from the baseline individual cooling equipment that is scrapped as a result of the project activity are accounted taking into consideration the decision taken by the Board at its 34th meeting, Paragraph 17 of the meeting report.

negligible, this source of leakage shall be considered once, i.e. at the first monitoring and calculated as below:

$$LE_{water} = Q_{water} \times EF_{water}$$ Equation (9)

Where:

LE_{water} = Emissions due to the freshwater usage in the project system (tCO2e)

Q_{water} = Maximum designed quantity of freshwater to be used in the project system (t)

EF_{water} = Emission factor associated with the production of freshwater (tCO2e/t)

PA01

Emissions from fresh water used			Source
Maximum designed quantity of freshwater to be used in the project system (Q_{water})	1620.5	m3	Design Report page 28 for 9000 TR project activity
Emission Factor (EF_{water})	11.5	kg/m3	Page 80 Taqa sustainability report 2022 https://www.taqa.com/wp-content/uploads/2023/03/20230315-TAQA-AR-2022_ENG_Full-V7.pdf
Emissions (LE_{water})	**19**	tCO2	This is applicable for first year only.

PA02

Emissions from fresh water used			Source
Maximum designed quantity of freshwater to be used in the project system (Q_{water})	2260	m3	Design Report page 4 for 9000 TR project activity
Emission Factor (EF_{water})	11.5	kg/m3	Page 80 Taqa sustainability report 2022 https://www.taqa.com/wp-content/uploads/2023/03/20230315-TAQA-AR-2022_ENG_Full-V7.pdf
Emissions (LE_{water})	**26**	tCO2	This is applicable for first year only.

The emission factor for water will be provided by water supplier Emirates Water and Electricity Company (EWEC) during the first monitoring. The publicly available value is used for estimation.

5.4 GHG Emission Reductions and Carbon Dioxide Removals

$$ER_y = BE_y - PE_y - LE_y$$ Equation (10)

Where:

ER_y	=	Emissions reductions in year y (t CO2e)
BE_y	=	Baseline emissions in year y (t CO2e)
PE_y	=	Project emissions in year y (t CO2e)
LE_y	=	Leakage emissions in year y (t CO2e)

For PA01

Vintage period	Baseline emissions (tCO_2e)	Project emissions (tCO_2e)	Leakage emissions (tCO_2e)	Reduction VCUs (tCO_2e)	Removal VCUs (tCO_2e)	Total VCUs (tCO_2e)
25-May-2023 to 31-Dec-2023	25,477	14,442	19	11,016	0	11,016
01-Jan-2024 to 31-Jul-2024	20,556	11,019	0	9,537	0	9,537
Total	**46,033**	**25,461**	**19**	**20,553**	0	**20,553**

For PA02

Vintage period	Baseline emissions (tCO_2e)	Project emissions (tCO_2e)	Leakage emissions (tCO_2e)	Reduction VCUs (tCO_2e)	Removal VCUs (tCO_2e)	Total VCUs (tCO_2e)
25-May-2023 to 31-Dec-2023	0	0	0	*0*	0	0
01-Jan-2024 to 31-Jul-2024	8,238	5,937	26	2,275	0	2,275
Total	**8,238**	**5,937**	**26**	**2,275**	**0**	**2,275**

Total

Vintage period	Baseline emissions (tCO_2e)	Project emissions (tCO_2e)	Leakage emissions (tCO_2e)	Reduction VCUs (tCO_2e)	Removal VCUs (tCO_2e)	Total VCUs (tCO_2e)
25-May-2023 to 31--2023	25,477	14,442	19	11,016	0	11,016
01-Jan-2024 to 31-Jul-2024	28,794	16,956	26	11,812	0	11,812
Total	**54,271**	**31,398**	**45**	**22,828**	**0**	**22,828**

For projects required to assess permanence risk:

Not applicable

PA01

Vintage period	Ex-ante estimated reductions/ removals	Achieved reductions/ removals	Percent difference	Explanation for the difference
PA01 25-May-2023 to 31-Dec-2023	11,681	11,016	-5.69%	Its negative difference due to less generation
PA01 01-Jan-2024 to 31-Jul-2024	11,272	9,537	-15.39%	Ex-ante ER 19323 is apportioned for 7 months (11272). The ER is reduced due to lower cooling production.
Total	22,953	20,553	-10.46%	

PA02

Vintage period	Ex-ante estimated reductions/ removals	Achieved reductions/ removals	Percent difference	Explanation for the difference
PA02 25-May-2023 to 31-Dec-2023	0	0	0	The PA02 started on 3rd May 2024
PA02 01-Jan-2024 to 31-Jul-2024	4,646	2,275	-51.03%	Ex-ante ER 19323 is apportioned for 88 days (4646). The ER is reduced due to lower cooling production.
Total	4,646	2,275	-51.03%	

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Not Applicable